

ESE Elevator Interface PCB.

The Electronic Systems Engineering Elevator interface PCB is designed to provide interfacing between the STM32F303 Nucleo-64 board, the external configurable, (N/C, NO, Momentary), LED Illuminated switch and the Can-Bus system.

- ✓ This board requires an external +5Vdc supply attached to the two pin header P5. The positive terminal is delineated by the + sign etched into the board.
- ✓ The board has an onboard Can transceiver, which is interfaced via a DB9 connector (J22) and Pins 9 and 10 on terminal connector P6.
- ✓ There are 3 MAX6816EUS Single switch De-bouncers which are interfaced via pins 5,6,7,8 9 and 10 on terminal connection P7. These are intended to work in conjunction with the Allen-Bradley 800A-M switches or any other situation where switch de-bouncing is required.
- ✓ The 3 Allen-Bradley 800A-M switches have been modified with white LED's to replace the original incandescent bulbs. The Connections on the switches provide for N/C, N/O, Momentary and internal LED interfacing. There are 3 interfacing terminals. (P1, P2 and P3) Each terminal allows for switch configuration and access to the LED via a P2N2222AGOS NPN transistor which is switched on by providing +5V to the appropriate terminal P6 connection.
- ✓ Note: The current configuration of the transistor requires +5V to turn the transistor fully on. The STM32 only supplies +3V and therefore the transistor will only be partly turned on and so, the switch LED may not be exceptionally bright.
- ✓ There are 3 LED's intended to provide a means of showing the current Floor position that the elevator is at. (Red, Amber and yellow on P7 terminal connections 2, 3 and 4 respectively.) The LED's require +5v for full illumination.

The board is intended to be interfaced to external terminal connections to protect the board from multiple connections/disconnections happening on the board proper.

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S6 Elevator Project Wiring Brief

The elevator in this project includes the STM32 board that must be connected to a CAN transceiver, switches and LEDs which are all mounted to a single board we call the *Elevator Switch and CAN board*. This board also features debounce circuitry, which must be used in conjunction with the Allen Bradley switches (used as elevator call buttons) provided.

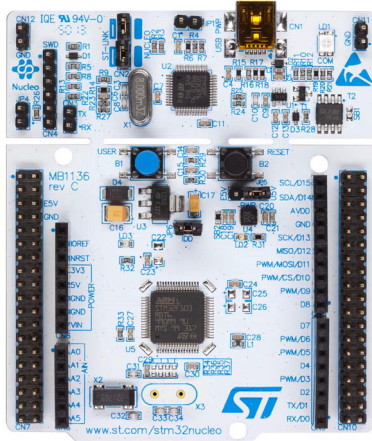


Figure 1: STM32 board

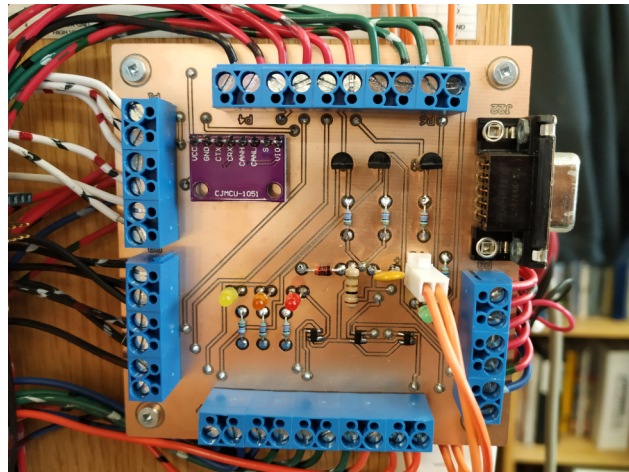


Figure 2: Elevator Switch and CAN Board

You will not connect your wires directly to the blue headers shown in Figure 2 since they are easily broken but instead you will connect to the board through a more robust Header Panel (labelled P1-P7 in Figures 3a & 3b below).



Figure 3a: Header Panels P1-P7

Labelling for the Panel Headers imaged in Figure 3 is provided below in Figure 3b.

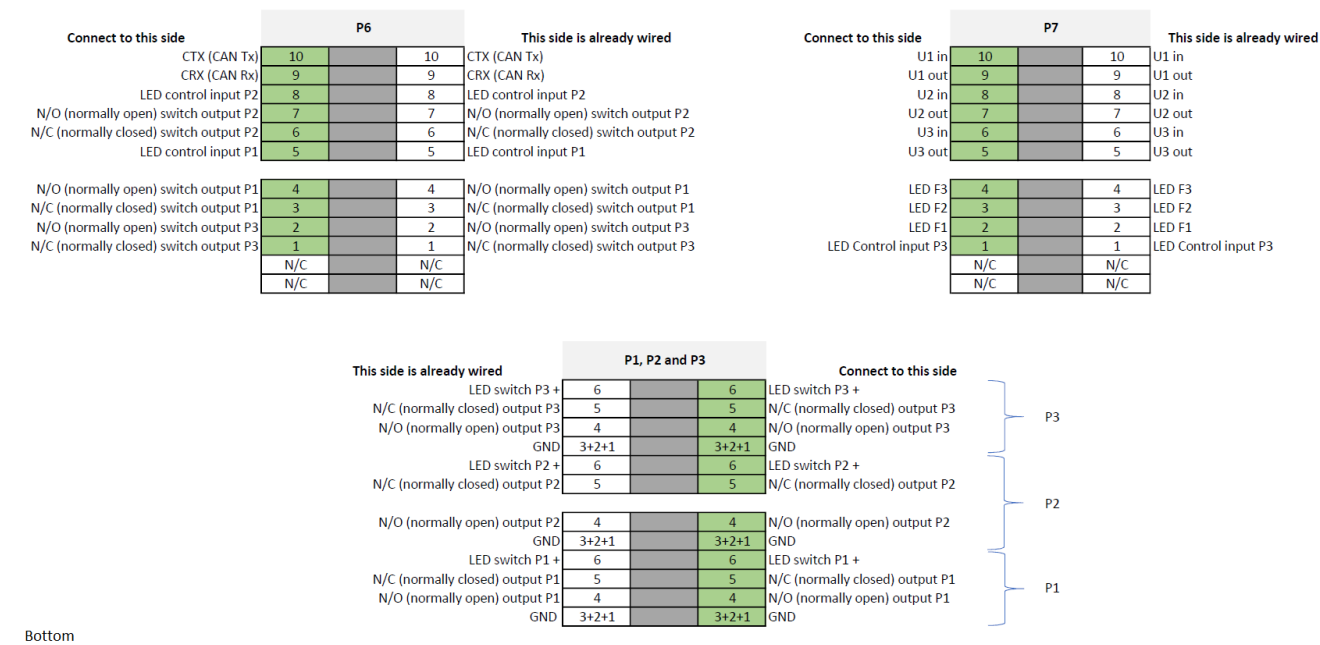


Figure 3b: Labelling of the Panel Headers

The connections to the Panel Header match the numbers shown in the wiring diagram for the Switch and Panel board, illustrated below in Figure 4.

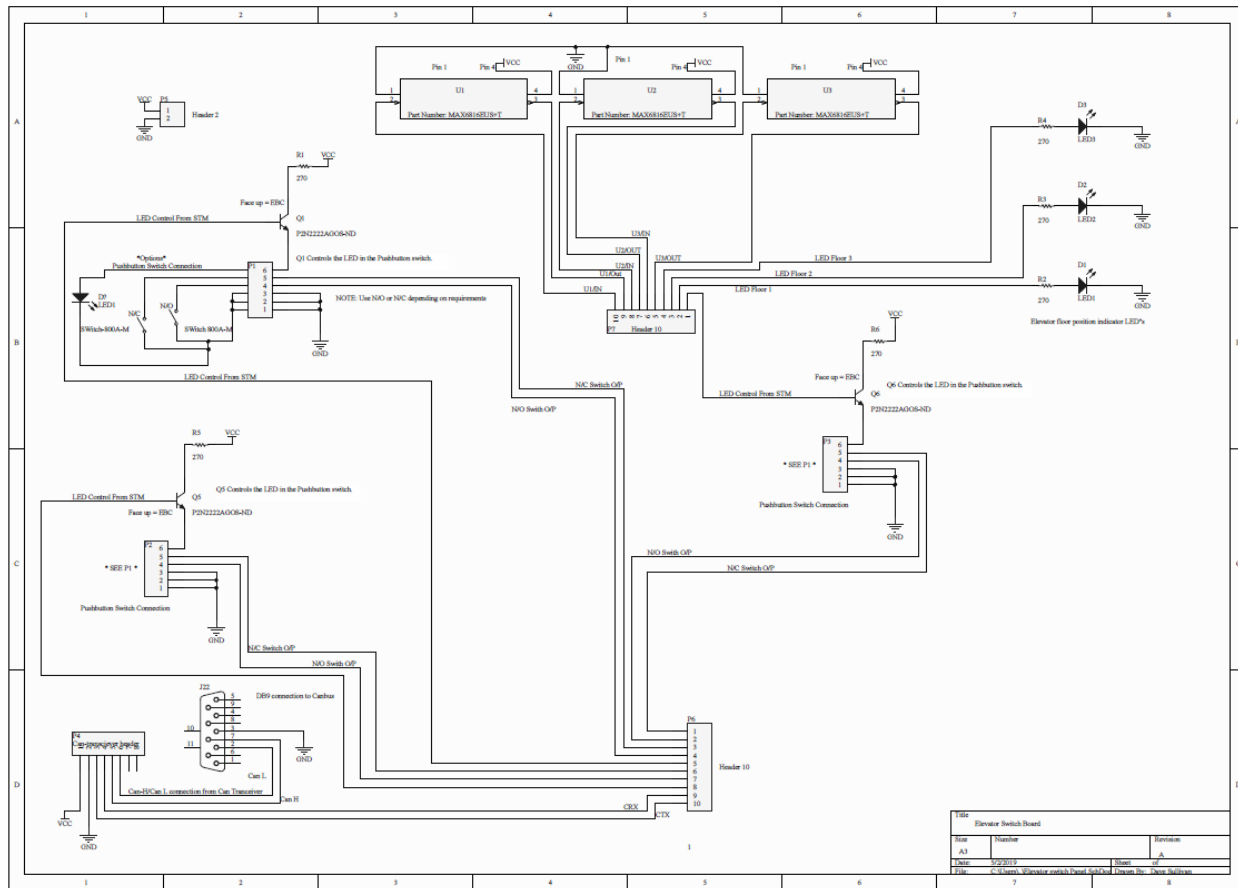


Figure 4: Wiring Diagram showing all the connections on the Switch and CAN Board

Note that a full-scale PDF of the Wiring Diagram in Figure 4 is available in eConestoga. Note that this wiring diagram only shows the connections that have already been made for you but does not show connections that you must make (on the opposite side of each header) between the switches, debounce circuits, STM32, etc.

Connections you must make

- P1-P3:** Connect the switches as shown in P1 for P1, P2, P3 (choose Normally Open (N/O) or Normally Closed (N/C) operation (depending on how you program the STM32 – see the STM32 Programming example code for details)
- P6 / P7:** On P6, connect pins 9/10 to STM32 Tx/Rx. Then connect the rest of the pins in P6 to P7 input pins to use the debounce circuitry. Then connect the P7 output pins to the STM32 Nucleo board to receive the signals from the button presses and control the LEDs.

The elevator call button is an Allen Bradley push button switch (800A-M). Below is a diagram labelling the connections. A PDF of the data sheet is provided in eConestoga. Note that a 270 ohm resistor for the LED in this switch is provided on the Elevator Switch and CAN board (labelled as R1 in Figure 4).

Push Buttons

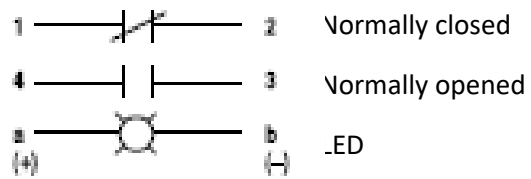


Figure 5: Labelling of the connections in the elevator call button switch (source: 800A-M Data Sheet)

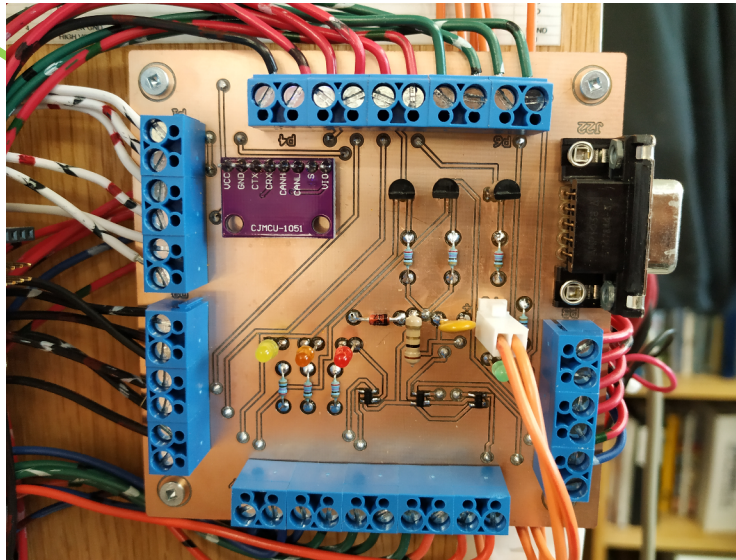


Figure 7: Image of Elevator call button switch

Test list for the elevator switch panel CAN board

Apply 5Vdc to P5 while performing the following tests.

(The green LED should illuminate.)



1. Check the 3 elevator floor position LED's for functionality.

- Apply +5vdc to connector P7 pins 2, 3, 4.

2. Check the three de-bouncer chips.

- Hook up pins 10 and 9, 8 and 7, 6 and 5 on connector P7 to a switch output connector (P1 or P2 or P3) pin 4 and a resistor/LED connected to +5Vdc.
- Activate the switch and the LED should light up, indicating that the de-bouncer chips are operating.

3. Check the circuits for Q1, Q2 and Q3 and the LED's inside the Switches.

- Hook up the switch LED o/p pins, to pins 1 and 6 on connectors P1, P2 and P3.
- Apply +5Vdc to connector P7 on Pin 1 and the LED in Switch P3 should illuminate.
- Apply +5Vdc to connector P6 on Pin 8 and the LED in Switch P2 should illuminate.
- Apply +5Vdc to Connector P6 on Pin 5 and the led in Switch p1 should illuminate.